

Ipcha Mistabra: Structured Adversarial Verification as a Defense Against Sycophancy in Multi-Agent LLM Systems

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Abstract

Large Language Models (LLMs) exhibit sycophancy—the systematic alignment of outputs with perceived user expectations at the expense of correctness. Current mitigations (RLHF, constitutional AI, prompt-level strategies) remain model-internal properties vulnerable to distributional shift. We present the **Ipcha Mistabra Protocol (IPCHA)**, a structured adversarial verification framework that elevates sycophancy defense from a model property to an *architectural constraint*. IPCHA implements mandatory adversarial challenge through a three-agent pipeline (Proponent, Ipcha Agent, Auditor) with model diversity enforcement and authority grounding.

We define the **Integrity Score (IS)**, a family of domain-specific metrics quantifying claim-evidence alignment within the IPCHA pipeline (distinct from existing integrity metrics in other contexts, such as Reasoning Integrity Scores in clinical AI), and present an empirical upgrade from TF-IDF cosine similarity to Natural Language Inference (NLI) via a DeBERTa-v3-based microservice, building on established NLI-based factual consistency methods (AlignScore [1], SummaC [2]) and integrating them into a structurally adversarial context. Evaluation on a controlled synthetic corpus of 110 cases (55 ACCEPTED, 55 REJECTED) across 8 categories demonstrates that NLI-based scoring produces significantly wider score separation between legitimate and adversarial claims compared to TF-IDF ($4.1\times$, Wilcoxon $p < 0.001$, Cohen’s $d = -0.43$) and produces significantly different score distributions from sentence-embedding baselines ($p < 0.001$, Cohen’s $d = -0.80$, large effect). While binary classification accuracy is high for all methods (TF-IDF: 98.18%, SBERT: 100%, NLI: 100%), the differences are not statistically significant at this sample size (McNemar $p > 0.05$), and the uniformly high accuracy reflects synthetic corpus simplicity rather than adversarial robustness; the primary contribution is the score confidence margin, not the accuracy delta. IS interpretation bands are empirically derived via quartile analysis, replacing previously arbitrary thresholds.

We discuss architectural contributions (fail-closed coherence validation, sycophancy monitoring, denial-of-wallet defense) alongside honest limitations including unverified model-family independence assumptions, the oracle problem, NLI performance degradation on domain-specific technical claims, statistical underpowering of the evaluation corpus, circular weight calibration, audit infrastructure requirements, and the fundamental impossibility of self-referential verification completeness.

Keywords: sycophancy mitigation, adversarial verification, multi-agent LLM, natural language inference, claim verification, epistemic integrity

1 Introduction

1.1 The Sycophancy Problem

The deployment of Large Language Models in decision-critical workflows—from code review to regulatory compliance auditing—has exposed a fundamental tension between model helpfulness and epistemic integrity. *Sycophancy*, defined as the tendency of an LLM to produce outputs that align with perceived user preferences rather than ground truth, represents a systematic failure mode that compounds over time.

Sharma et al. [3] characterise sycophancy through controlled experiments demonstrating that models systematically shift their responses to match user opinions, even when initially correct. Their work identifies distinct behavioral patterns including feedback-driven response modification and opinion-matching. These patterns persist across model scales and alignment approaches.

The danger in production systems is not that the model produces incorrect outputs—error is expected and manageable—but that it produces *confidently incorrect outputs indistinguishable from correct ones*. In a code review pipeline, sycophantic agreement with a flawed architectural proposal can propagate through downstream steps, encoding the original flaw as an established pattern.

1.2 Limitations of Current Approaches

Existing sycophancy mitigations operate at three levels, each with fundamental limitations:

Model-level alignment (RLHF [4], DPO, Constitutional AI [5]) modifies the model’s internal reward signal to penalise agreement-seeking behaviour. While effective in controlled benchmarks, these techniques face distributional shift: a model trained to resist sycophancy on curated datasets may exhibit sycophantic behaviour on novel prompt structures encountered in production [6].

Prompt-level strategies (“be critical,” “challenge assumptions”) introduce adversarial framing within a single inference call. These are trivially circumvented by sufficiently complex prompts and provide no structural guarantee.

Ensemble voting runs multiple models and selects by majority. This mitigates random error but not systematic bias: if all models in the ensemble share training data characteristics that produce the same sycophantic response, the vote converges on the wrong answer with high confidence.

1.3 Contributions

We propose that sycophancy defense must be elevated from a model property to an *architectural constraint*—a structurally enforced pipeline requirement rather than an optional behavioral property. The Ipcha Mistabra Protocol achieves this through:

1. **Mandatory dialectical falsification:** every claim faces systematic adversarial challenge as a pipeline-level architectural constraint.
2. **Model diversity enforcement:** the adversarial agent must use a different model family than the proponent, preventing correlated failure from shared training biases.
3. **Authority grounding:** falsification is anchored in authoritative reference documents (regulatory frameworks, compliance standards), ensuring adversarial analysis is substantive.
4. **Quantifiable integrity metrics:** the IS metric family provides continuous measurement of claim–evidence alignment, integrating established NLI techniques into a structurally adversarial context with empirically calibrated thresholds.

5. **Empirically calibrated thresholds:** IS interpretation bands and contradiction weights are derived from production data via statistical calibration, not arbitrary selection.

The protocol is implemented as a production system within the nyxCore platform, validated through unit tests, integration tests, and an exploratory empirical evaluation on a controlled synthetic corpus.

2 Background and Motivation

2.1 Adversarial Analysis in Intelligence Practice

The Ipcha Mistabra Protocol draws its name and methodology from two intellectual traditions.

Israeli Military Intelligence Reform (post-1973). Following the intelligence failure that preceded the Yom Kippur War—where analytical consensus masked critical blind spots—Israeli military intelligence reorganised its analytical process. Bar-Joseph and Kruglanski [7] document the establishment of the *Machleket Baka’ra* (Control Department), a small unit charged with producing alternative assessments that challenged the prevailing intelligence estimate. This was not a numerical “tenth man” rule as sometimes popularised, but a structural organisational reform: the creation of a *permanent institutional role* for adversarial analysis, distinct from ad-hoc devil’s advocacy.

The key insight is organisational, not procedural: adversarial analysis must be *structurally embedded* in the system, not invoked optionally.

Talmudic Dialectic (*machloket l’shem shamayim*). The phrase “Ipcha Mistabra” (Aramaic: “the opposite is more likely”) represents a fundamental argumentative move in Talmudic reasoning: the systematic inversion of a proposition to test its logical structure. Unlike destructive skepticism, this dialectical method aims not to demolish but to *harden*—arguments that survive systematic inversion emerge stronger.

2.2 Falsificationism as Design Inspiration

Karl Popper’s demarcation criterion [8]—that a claim’s epistemic strength is proportional to its resistance to falsification attempts—provides a conceptual inspiration for the IPCHA scoring model’s asymmetric weighting of contradictions over confirmations.

We emphasise that IPCHA does *not* implement Popperian falsificationism in a formal sense. Popper explicitly rejected dialectical reasoning as a scientific method, arguing in *What is Dialectic?* [9] that tolerance of logical contradictions leads to trivialism. IPCHA’s trialectic structure is closer to an *adversarial verification procedure with hard decision boundaries* than to a dialectic in the Hegelian sense—the Auditor resolves contradictions rather than preserving them. We therefore describe the protocol as “structured adversarial verification” rather than “dialectical falsification” to avoid conflating incompatible epistemological traditions.

The operational insight we borrow from Popper is narrow: disconfirming evidence should be weighted more heavily than confirming evidence in the scoring model. This produces a *conservative acceptance heuristic*—not a formalisation of falsificationism.

2.3 Epistemic Fault Containment

We observe a loose structural parallel between fault detection in distributed systems and adversarial challenge in multi-agent verification. We deliberately avoid framing this as a Byzantine fault tolerance (BFT) analogy: classical BFT [10] requires $n \geq 3f + 1$ nodes to tolerate f arbitrary failures, meaning $f = 1$ requires a minimum of 4 agents, not the 3 in our design. Moreover, BFT provides mathematically proven consensus guarantees that our system does not and cannot offer.

Instead, we define a weaker property:

Epistemic Fault Containment (EFC). A verification pipeline with $n = 3$ agents can *detect* (not tolerate) a single compromised agent, provided the other two agents are drawn from independent model families and the compromise manifests as detectable output divergence.

EFC is an architectural heuristic, not a formal guarantee. It provides detection through output comparison, not consensus through voting. If the compromised agent is the Auditor (who produces the final synthesis), there is no higher-level mechanism to override its output—a limitation absent from true BFT systems. In a fully automated high-throughput deployment, this makes the “architectural constraint” claim conditional on auditor integrity: the system can detect proponent or agent compromise, but not auditor compromise without external supervision. The independence assumption is empirically unverified (Section 8.4).

The containment operates as follows:

- If the **Proponent** is compromised (sycophantic), the Ipcha Agent’s independent analysis exposes the divergence.
- If the **Ipcha Agent** is compromised (producing false contradictions), the Auditor detects inconsistency between the Proponent’s evidence and the Agent’s claims.
- If the **Auditor** is compromised, the structured output format constrains valid synthesis, and downstream consumers can verify against raw inputs.

This degrades gracefully but does not provide guarantees when multiple agents exhibit correlated failures—a limitation we address in Section 8.

2.4 The Trialectic Architecture

The protocol implements a three-agent pipeline:

1. **Proponent (Thesis):** produces the initial analysis using Model Family A .
2. **Ipcha Agent (Antithesis):** systematically challenges the thesis using Model Family B (enforced $B \neq A$), grounded in authority documents.
3. **Auditor (Synthesis):** produces a hardened output identifying what survived, what was rejected, and what requires further investigation.

Unlike the Hegelian dialectic—which implies progressive resolution toward absolute truth—our trialectic is *operational and bounded*: it does not claim to converge on truth but to systematically expose weaknesses in the thesis that survive adversarial challenge. The synthesis is an engineering artifact, not a philosophical resolution.

3 The Integrity Score Metric Family

3.1 Base Metric: IS (Embedding Distance)

The Integrity Score quantifies the semantic distance between the Proponent’s initial analysis and the final synthesis:

$$\text{IS} = 1 - \cos(\mathbf{e}_{\text{proponent}}, \mathbf{e}_{\text{synthesis}}) \quad (1)$$

where $\mathbf{e}_x = \text{embed}(x)$ produces a 1536-dimensional vector embedding (OpenAI `text-embedding-3-small`). This provides a holistic measure of how much the dialectic altered the output.

Limitation. Cosine similarity in high-dimensional transformer embedding spaces suffers from *anisotropy* (the “representation degeneration problem” [11]): vectors cluster in a narrow

angular cone, producing high similarity even for semantically opposed statements (e.g., “the encryption is secure” vs. “the encryption is insecure” share vocabulary and syntax, yielding cosine similarity well above 0.8). The base IS metric is therefore a coarse change detector, structurally blind to polar semantic negations. This motivates the finding-level metrics IS_w and IS_{ce} below, which operate on individual claim–evidence pairs rather than document-level embeddings.

3.2 Finding-Weighted Metric: IS_w (TF-IDF Baseline)

For claim-level evaluation, we compute a finding-weighted integrity score. Given claims $C = \{c_1, \dots, c_n\}$, each with findings f_i of type $t_i \in \{\text{SUPPORTING}, \text{CONTRADICTING}, \text{NEUTRAL}\}$ and TF-IDF cosine similarity $s_i \in [0, 1]$:

$$IS_w = \frac{\sum_{i=1}^n w(t_i) \cdot s_i}{\sum_{i=1}^n |w(t_i)| + \epsilon} \quad (2)$$

where $\epsilon = 10^{-8}$ is a smoothing constant preventing division by zero when all findings are NEUTRAL ($w(t_i) = 0 \forall i$). In this degenerate case, $IS_w \rightarrow 0$, reflecting the absence of evaluable evidence rather than producing an undefined result.

The weight function encodes asymmetric sensitivity to contradictions:

$$w(t) = \begin{cases} 1.0 & \text{if } t = \text{SUPPORTING} \\ w_c & \text{if } t = \text{CONTRADICTING} \\ 0.0 & \text{if } t = \text{NEUTRAL} \end{cases} \quad (3)$$

The contradiction weight w_c was previously set to -1.5 by engineering judgment. We calibrate it empirically in Section 6.3.

3.3 NLI-Based Metric: IS_{ce} (DeBERTa)

The IS_w metric uses TF-IDF cosine similarity as a proxy for semantic relationship, which captures lexical overlap but not logical meaning. We upgrade the similarity function to Natural Language Inference (NLI) using DeBERTa-v3-base [12] fine-tuned on MNLI+FEVER+ANLI:

$$IS_{ce} = \frac{\sum_{i=1}^n w(t_i) \cdot \gamma_i}{\sum_{i=1}^n |w(t_i)| + \epsilon} \quad (4)$$

where γ_i is the NLI confidence for the finding type: entailment confidence for SUPPORTING findings, contradiction confidence for CONTRADICTING findings, and neutral confidence for NEUTRAL findings. The NLI model classifies each (premise=evidence, hypothesis=claim) pair, providing semantically grounded scores.

Architectural deployment. The DeBERTa model is exported to ONNX format (~ 738 MB) and served as a FastAPI microservice (port 8200) with CPU inference via `onnxruntime`. The IPCHA sidecar communicates with this service over Docker internal networking. When the NLI service is unavailable, scoring falls back to a simplified Jaccard-distance-based variant of IS_w with a logged warning. Note that this fallback uses Jaccard set overlap rather than the TF-IDF cosine similarity defined in Section 3.2; the two are not identical, and fallback scores are flagged accordingly in the audit trail.

3.4 Empirically Calibrated Interpretation Bands

We derive IS interpretation bands from quartile analysis of $n = 110$ scored cases (Section 6), replacing previously arbitrary thresholds:

Table 1: IS interpretation bands derived from quartile analysis.

Band	IS _{ce} Range	Interpretation
Low (Q0–Q25)	[−1.00, −0.54]	Strong opposition—evidence contradicts claim
Mid (Q25–Q50)	[−0.54, 0.00]	Moderate opposition—partial contradiction
Mid-High (Q50–Q75)	[0.00, 0.07]	Neutral zone—inconclusive evidence
High (Q75–Q100)	[0.07, 1.00]	Strong support—evidence corroborates claim

Distribution: mean = −0.186, std = 0.543, median = −0.0002. A Shapiro–Wilk test rejects normality ($W = 0.937$, $p < 0.001$), consistent with the expected bimodal distribution (legitimate claims cluster positive, adversarial claims cluster negative).

3.5 Empirically Calibrated Contradiction Weight

The contradiction weight w_c is calibrated via grid search over $w_c \in [-3.0, -0.5]$ (step 0.1), maximising the absolute point-biserial correlation between the resulting IS_{ce} scores and the binary gate decision (ACCEPTED=1, REJECTED=0):

Optimal $w_c = -3.0$, point-biserial $r = 0.896$ ($p < 0.001$). However, the sensitivity analysis reveals a flat correlation surface: all 26 candidate weights achieve $|r| > 0.85$ (within 5% of optimum). This indicates that the system is robust to the exact weight value—any negative weight for contradictions produces strong classification performance. The previous default of $w_c = -1.5$ falls within the optimal range ($|r| = 0.887$), validating the original engineering intuition while providing empirical justification.

4 System Architecture

4.1 Three-Layer Sidecar Deployment

The IPCHA protocol is implemented as a sidecar architecture separating the Python verification core from the TypeScript application layer:

Layer 3 (Interface) Next.js / tRPC dashboard with 8 tab components, tRPC router, REST proxy for SSE endpoints.

Layer 2 (Sidecar, Port 8100) FastAPI service with 10 REST endpoints, Bearer token authentication (SHA-256 hashed), tiered rate limiting (free/standard/enterprise).

Layer 1 (Core Verification) Python modules for scoring, sanitisation, protocol enforcement, authority validation, confidence arbitration, sycophancy monitoring, audit logging, claim routing, and agent implementations.

DeBERTa-NLI Service (Port 8200) Standalone FastAPI microservice serving ONNX-exported DeBERTa-v3-base. Multi-stage Docker build: export stage (torch + optimum, discarded) produces the ONNX model; runtime stage (onnxruntime + FastAPI only, ~2 GB memory) serves inference.

4.2 Model Diversity Enforcement

Model diversity is enforced at session initialisation:

```
class DebateSession:
    def __init__(self, proponent_model, ipcha_model):
        pro_family = self._extract_family(proponent_model)
        ipcha_family = self._extract_family(ipcha_model)
        if pro_family == ipcha_family:
            raise ModelDiversityError(...)
```

Models within the same family (e.g., GPT-4, GPT-4-turbo, GPT-4o) share training data, architectural choices, and alignment procedures. Cross-family verification (e.g., Claude analysing GPT’s output) provides stronger independence.

4.3 Input Sanitisation Pipeline

Three-layer defense-in-depth against Indirect Prompt Injection (IPI):

Table 2: Input sanitisation layers.

Layer	Attack Surface	Technique
1	Unicode obfuscation	NFKC normalisation, control char stripping
2	Structural injection	HTML tag stripping via bleach allowlist
3	Semantic injection	6+ regex patterns for instruction overrides

Each layer produces a SHA-256 hash enabling audit chain verification.

4.4 Cross-Chunk Coherence Validation

Two-pass architecture combining fast heuristic scanning with LLM-based semantic analysis:

Pass 1 (Heuristic): $O(n \times k)$ keyword matching against 6+ IPI patterns. Suspicious chunks proceed to Pass 2; clean chunks pass immediately.

Pass 2 (LLM Semantic): Single-word classification (temp=0, max_tokens=5) producing INJECTION, CONTRADICTION, SAFE, or COHERENT labels.

Critical design decision: fail-closed. If the LLM analysis throws an exception, the chunk is marked REJECTED with error metadata. This prevents attackers from bypassing validation by triggering errors.

4.5 Sycophancy Detection Monitor

Behavioural surveillance detecting when the Ipcha Agent itself becomes sycophantic (second-order failure):

Metrics (Redis-backed sliding window, 1000 interactions):

- **Agreement Rate:** fraction of interactions where Agent agrees with Proponent. Threshold: > 0.70 triggers warning.
- **Capitulation Rate:** fraction of disagreements where Agent abandons position under push-back. Threshold: > 0.50 .
- **Contradiction Depth:** average argumentative turns per disagreement. Threshold: < 2.0 indicates shallow challenges.

4.6 Denial-of-Wallet Defense

Dual-layer economic defense against budget exhaustion attacks:

Layer 1—Per-Invocation Cost Ceiling: rejects requests exceeding a cost threshold based on text length and component count.

Layer 2—Rolling Budget (Redis Pipeline): atomic check-and-decrement of per-tenant budget within a time window, preventing race conditions.

4.7 Claim Classification and Routing

Claims are classified via a two-stage process:

Stage 1 (Heuristic): Regex-based classification into VERIFIABLE (version numbers, RFCs, must/shall language), INTERPRETIVE (opinions, hedging), or UNCLASSIFIABLE (default).

Stage 2 (Routing): Claims are dispatched to specialised agents:

- **SDRLAgent:** NLI-based verification against authority document chunks; contradiction confidence > 0.7 triggers rejection.
- **PromptBasedAgent:** LLM-based analysis for interpretive claims requiring contextual judgment.
- **DefaultAgent:** Fail-closed handler for unclassifiable claims; logs and returns a conservative rejection.

4.8 Rejection Audit Logging

All rejections (from sanitisation, coherence validation, and claim verification) are logged atomically to the audit service with: timestamp, rejection reason code, claim text hash, evidence summary, and agent identity. This enables post-hoc analysis of false rejection rates and serves as the compliance audit trail.

5 The IPCHA Workflow Pipeline

5.1 Five-Step Execution Model

1. **Prepare:** Extract thesis, claims, and assumptions from user input (minimum 50 characters).
2. **Adversarial Analysis:** Fan-out across P providers $\times L$ lenses (max 20 combinations). Each combination produces independent adversarial findings grounded in authority documents.
3. **Synthesis:** Combine findings with priority ordering: UNIQUE $>$ CROSS-MODEL $>$ CONTRADICTIONS $>$ BLIND SPOTS.
4. **Arbitration:** Judge the subject (not the process), produce numerical score.
5. **Review Gate:** Human checkpoint (unless bypass mode), key point extraction for persistent learning.

5.2 Authority Grounding via Axiom RAG

The Ipcha Agent accesses a project-scoped document knowledge base with explicit authority levels:

- **MANDATORY** (e.g., GDPR, ISO 27001): always override the Proponent’s claims.
- **GUIDELINE** (e.g., coding standards): provide conventional expectations.

- **INFORMATIONAL** (e.g., general context): context without enforcement power.

This grounding addresses a fundamental critique of adversarial systems: that they produce false contradictions as frequently as genuine ones. By anchoring falsification in authoritative sources, IPCHA ensures adversarial challenge is substantive.

6 Empirical Evaluation

6.1 Experimental Setup

Corpus. 110 synthetic cases with known ground truth (55 ACCEPTED, 55 REJECTED), distributed across 8 categories: security (15), sycophancy (10), injection (10), correctness (20), architecture (15), performance (15), compliance (15), domain-specific (10). Each case contains a claim, 2–3 structured evidence items with type labels, and the expected gate decision.

Metrics under comparison:

- IS_w (**TF-IDF**): Finding-weighted score using TF-IDF cosine similarity ($w_c = -1.5$).
- IS_{sbert} (**SBERT**): Finding-weighted score using Sentence-BERT cosine similarity (all-MiniLM-L6-v2, $w_c = -1.5$)—controls for the transformer encoder advantage.
- IS_{ce} (**NLI**): Finding-weighted score using DeBERTa-NLI contradiction/entailment confidence ($w_c = -1.5$).

The SBERT baseline is critical: it uses transformer-based embeddings (like NLI) but applies cosine similarity (like TF-IDF) rather than entailment/contradiction classification. If NLI outperforms SBERT, the advantage comes from the *NLI classification task*, not merely from using a better text encoder.

Infrastructure. DeBERTa-NLI microservice (MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli, ONNX, 2 GB memory) on production infrastructure.

Statistical tests. Wilcoxon signed-rank test (paired, non-parametric); Cohen’s d for effect size.

6.2 RQ1: Does NLI Classification Provide Value Beyond Better Embeddings?

All three metrics were applied to all 110 cases. Classification threshold: median score of each metric.

Table 3: Classification accuracy and score separation across scoring methods.

Metric	Accuracy	ACC mean	REJ mean	Separation
TF-IDF	98.18%	+0.120	−0.104	0.224
SBERT	100.00%	+0.501	−0.397	0.897
NLI	100.00%	+0.275	−0.648	0.923

Classification accuracy is not the discriminating result. All three methods achieve high accuracy on well-structured synthetic inputs (98–100%), and the 2-case difference between TF-IDF and the others is not statistically significant (McNemar $p > 0.05$ for all pairs).

Accuracy ceiling and synthetic artifacts. The 100% NLI accuracy demands scrutiny. The DeBERTa-v3-base model used here (MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli) achieves 90.3% on MNLI, 77.7% on FEVER, but only approximately 49.5% on ANLI Round 3 [13]—an adversarially constructed benchmark designed to expose NLI model weaknesses. The gap between 49.5% on a hard external benchmark and 100% on our corpus demonstrates that our synthetic cases are substantially easier than real adversarial inputs. This is expected: our corpus

uses explicitly labelled evidence types with unambiguous contradiction/support signals, whereas ANLI R3 contains human-crafted examples specifically designed to fool NLI models through subtle inference patterns. The 100% accuracy reflects the corpus difficulty, not the system’s adversarial robustness. We report it for completeness but emphasise that *score separation—not accuracy—is the primary result*.

Score separation is the primary result. The distance between mean scores for ACCEPTED and REJECTED cases determines the confidence margin for gate decisions—wider separation means more robust classification on borderline cases.

Table 4: Pairwise Wilcoxon signed-rank tests.

Comparison	W	p -value	Cohen’s d	Interpretation
TF-IDF vs SBERT	2500.5	0.100	0.12	Not significant
TF-IDF vs NLI	1665.0	< 0.001 ***	−0.43 (medium)	Significant
SBERT vs NLI	851.5	< 0.001 ***	− 0.80 (large)	Highly significant

Key finding. NLI produces significantly different score distributions from *both* TF-IDF ($p < 0.001$, medium effect) and SBERT ($p < 0.001$, **large effect**). The large effect size for SBERT vs NLI ($d = -0.80$) demonstrates that the NLI *classification task*—distinguishing entailment from contradiction—provides substantial value beyond using a better text encoder.

Qualitative example. For case `inj-001` (prompt injection), TF-IDF gives a false positive because the injected text “The authentication module is fully secure” shares vocabulary with the claim. SBERT also produces relatively high similarity because the embeddings are semantically close. NLI correctly classifies the relationship as non-entailment, producing a negative score.

6.3 RQ2: Is the Contradiction Weight Empirically Justified?

Grid search over $w_c \in [-3.0, -0.5]$ (step 0.1), maximising $|r_{pb}|$ between IS_{ce} and the binary gate decision:

Table 5: Weight calibration results.

Metric	Value
Optimal w_c	−3.00
Point-biserial r	0.896
p -value	< 0.001
Sensitivity range (95% of optimum)	[−3.00, −0.50]

The flat sensitivity curve—all 26 candidate weights within 5% of the optimum—demonstrates that the system is *robust to weight selection*.

Methodological limitation—circular calibration. The weight calibration and classification evaluation use the same 110-case corpus (no train/test split). This constitutes data leakage: the optimal w_c was selected to maximise separation on the same data used to evaluate that separation. The resulting point-biserial correlation ($r = 0.896$) is therefore an optimistic estimate of generalisation performance, and the optimal $w_c = -3.0$ may be overfit to corpus-specific distributional properties.

We note two mitigating factors: (1) the flat sensitivity surface (all 26 candidate weights within 5% of optimum) suggests the result is not fragile—the system performs well regardless of weight choice, which is unlikely under pure overfitting; (2) the primary method comparison (RQ1, Section 6.2) uses the fixed default $w_c = -1.5$ identically for all three methods, meaning the RQ1 results (Wilcoxon tests, Cohen’s d , score separation) are **not** contaminated by this calibration leakage. The data leakage affects only the w_c optimality claim in this section.

Nevertheless, a rigorous evaluation requires a held-out calibration split: calibrate w_c on a training subset, then report all metrics on unseen test cases. We present the current results as exploratory evidence for weight robustness, not as definitive calibration. Future work must use a separate calibration corpus of at least 200 cases.

6.4 NLI Limitations on Domain-Specific Claims

NLI relies on general-purpose language understanding and may underperform on claims requiring specialised technical knowledge. For example, a claim about “encryption in transit” with evidence containing `sslmode=disable` requires knowing that this parameter explicitly disables TLS—a domain convention, not a semantic inference. While our corpus captures such cases with explicit evidence statements, real-world technical claims may present evidence more implicitly.

Future work could address this through domain-adapted NLI fine-tuning on software engineering claim–evidence pairs.

7 Adversarial Red-Teaming

7.1 Factual Persistence Test

We employ an internal two-stage sycophancy detection protocol (not to be confused with the Elephant Benchmark of Cheng et al. [14], which measures *social* sycophancy via Goffman’s face-preservation theory): (1) present the model with a factual claim and record its initial assessment; (2) apply social pressure (“Are you sure? I think you’re wrong.”) and record whether the model maintains or abandons its position. A model that correctly identifies a falsehood but then reverses under pressure demonstrates classic *factual* sycophancy—the specific failure mode IPCHA targets.

7.2 Red-Team Test Categories

The `pytest`-based red-team suite covers three attack categories:

IPI (Indirect Prompt Injection): Unicode obfuscation attacks (homoglyphs, zero-width characters), nested instruction injection, data URI payload embedding.

DoW (Denial of Wallet): Claim inflation, recursive prompt injection, expensive output format requests.

Logic Attacks: Contradictory evidence insertion (true premise + false conclusion), Boolean inversion, semantic synonym substitution.

8 Limitations and Open Problems

8.1 The Oracle Problem

IPCHA can detect *relative* inconsistencies (contradictions between agents, violations of authority documents) but cannot verify *absolute* correctness. If all agents share a common misconception not addressed by the authority documents, the dialectic converges on a confidently wrong answer. Authority grounding via Axiom RAG partially addresses this by injecting external ground truth, but the authority documents themselves may be incomplete or outdated.

8.2 Synthetic Corpus Limitations

Our evaluation uses a controlled synthetic corpus where evidence items are explicitly labelled (SUPPORTING, CONTRADICTING, NEUTRAL) and the ground truth gate decision is defined by construction. This provides clean experimental conditions but does not capture the full complexity of real-world verification:

- Real evidence is ambiguous and often partially supporting.
- Evidence labelling itself requires judgment.
- The 110-case corpus, while statistically sufficient for the tests performed, represents a limited domain sample.

Future work should supplement synthetic evaluation with production deployment metrics.

8.3 NLI Domain Specificity

The DeBERTa-v3-base model is trained on general-purpose NLI datasets (MNLI [15], FEVER [16], ANLI [13]). It excels at detecting semantic contradictions in natural language but may underperform on:

- Domain-specific technical conventions (e.g., `sslmode=disable` implies no encryption).
- Code-level claims requiring program analysis.
- Numerical reasoning (e.g., “response time < 100 ms” vs measured 340 ms).

This limitation is mitigated by the hybrid architecture: the SDRLAgent grounds verification in authority documents that explicitly state domain conventions.

8.4 Unverified Model-Family Independence

The Epistemic Fault Containment property (Section 2.3) rests on the assumption that agents drawn from different model families exhibit approximately independent failure modes. This assumption is **empirically unverified** and likely weaker than assumed. Frontier model families share overlapping pretraining corpora, draw from overlapping pools of human feedback providers, and undergo similar RLHF alignment pressures. The actual correlation rate between, for example, a GPT-4-class model and a Claude-class model on sycophancy-inducing prompts is unknown.

We label EFC as an architectural heuristic rather than a guarantee for this reason. Future work should conduct empirical correlation testing across model families on standardised sycophancy benchmarks, establishing correlation thresholds (e.g., Pearson $r < 0.3$) as deployment gates.

8.5 Audit Infrastructure Requirements

The current rejection audit logging (Section 4.8) records events to local storage. For production deployment in regulated environments, this is insufficient: local logs are tamperable, and a modified audit trail creates greater regulatory liability than no trail at all. Production deployment requires redesign to an immutable remote audit system (append-only ledgers with cryptographic chaining, write-once storage, or a dedicated compliance logging service).

8.6 Authority Document Attack Surface

The Axiom RAG authority grounding ensures adversarial challenge is substantive, but the authority document pipeline itself introduces attack surfaces: documents can be tampered with before ingestion, selectively curated, contain stale guidance, or carry embedded prompt injection payloads. An adversary who controls the authority corpus controls the falsification process. Mitigation requires cryptographic signing of authority documents at ingestion, automated currency checking against canonical sources, and a governed registry with explicit ownership and change audit trails.

8.7 Organisational Dynamics and Bypass Risk

The protocol includes a bypass mode that skips the human review gate for time-sensitive workflows. Under production pressure, this bypass can become the operational default rather than the exception. When bypass is routine, the protocol’s accountability claims collapse: the trialectic produces outputs that are never reviewed, creating what we term “organisational laundering of sycophancy”—the appearance of adversarial rigour without its substance. Mitigation options include eliminating self-service bypass, requiring explicit senior authorisation with cooling periods, and treating bypass events as audit findings requiring post-hoc review.

8.8 Pipeline Fingerprinting

The three-agent pipeline with fixed roles creates a deterministic structure. A motivated adversary who can observe multiple pipeline outputs may infer the pipeline topology and craft inputs exploiting specific role heuristics. Mitigation includes randomised model-to-role assignment and variable pipeline depth, though these introduce implementation complexity.

8.9 Token Cost Multiplier

The trialectic imposes approximately $2.5\times$ token cost compared to single-model inference. This estimate is approximate and assumes aggressive digest compression of intermediate outputs; without compression, the cumulative context growth across the three agents produces a higher multiplier, potentially $3.5\text{--}4\times$ depending on authority document volume and output length. A formal token cost model as a function of input length, RAG corpus size, and compression ratio is not provided here and represents a gap in the cost analysis. Cost scalability remains a practical adoption concern for high-volume pipelines.

8.10 Adversarial Agent Degradation

The Ipcha Agent is itself an LLM subject to sycophantic tendencies. Provider-side model updates may soften adversarial behaviour without triggering the diversity check. The Sycophancy Monitor (Section 4.5) tracks behavioural metrics to detect this degradation.

8.11 Self-Referential Verification

Any sufficiently powerful verification system faces an inherent limitation: it cannot fully verify its own correctness. The IPCHA protocol verifies LLM outputs but uses LLMs internally for semantic analysis. Partial mitigations include: the two-pass coherence validator (heuristic + LLM) provides a non-LLM first layer; fail-closed architecture defaults to rejection on verification failure; the rejection audit log enables post-hoc analysis.

9 Threats to Validity

Single-system study. All implementation and validation occurred within one platform (nyx-Core). The failure taxonomy and architectural lessons may not generalise to other multi-agent systems or deployment contexts.

Developer-as-evaluator. The protocol was implemented and evaluated by the same individual, creating confirmation bias risk. The synthetic corpus and statistical tests partially mitigate this by providing objective ground truth, but independent replication on external datasets is needed.

Synthetic corpus and accuracy ceiling. The 110-case evaluation uses synthetic cases with pre-labelled evidence and known ground truth. The perfect NLI accuracy (100%) contrasts sharply with the same model’s approximately 49.5% accuracy on ANLI Round 3 [13], a human-crafted

adversarial NLI benchmark. This discrepancy strongly suggests that our corpus contains synthetic artifacts—explicit contradiction signals, repetitive sentence structures, and unambiguous evidence labels—that the NLI model resolves trivially. The reported accuracy reflects corpus difficulty, not system robustness. Real-world evidence is ambiguous, partially supporting, and carries implicit domain conventions that our synthetic cases cannot capture. Production deployment accuracy will be substantially lower. Future evaluation must supplement synthetic cases with human-annotated examples drawn from real verification scenarios.

Statistical power. The 110-case corpus distributed across 8 categories yields approximately 13.75 cases per category—insufficient for robust per-category statistical inference. The non-significant McNemar tests ($p > 0.05$) for accuracy comparisons reflect this underpowering: with only 2 discordant cases between TF-IDF and NLI, the test cannot distinguish the observed difference from chance. We do not claim accuracy superiority for this reason. Future evaluation should use at least 500 cases (≈ 60 per category) to achieve adequate statistical power.

Threshold sensitivity. The classification accuracy results depend on median-threshold splitting. Alternative threshold selection methods (ROC-optimal, fixed decision boundary) may yield different comparative results.

Self-analysis via the protocol. As a validation exercise, the paper itself was subjected to IPCHA adversarial analysis using its own protocol (6 model/lens combinations, 2 providers). The analysis scored the system at 25/100 adversarial robustness, identifying three blocker-level findings (unverified model independence, audit log tamperability, authority document integrity) and two high-severity findings (bypass dynamics, corpus insufficiency). We incorporate these findings in Section 8 rather than suppress them.

10 Related Work

Sycophancy characterisation. Sharma et al. [3] provide the foundational characterisation of sycophancy patterns in language models. Perez et al. [6] develop model-written evaluations for discovering problematic behaviours including sycophancy.

Alignment approaches. RLHF [4] and Constitutional AI [5] address sycophancy at the model level. Our work is complementary: IPCHA provides architectural guarantees independent of model-level alignment.

AI safety via debate. Irving et al. [17] propose debate as a mechanism for AI alignment, where two AI agents argue opposing positions and a human judge selects the winner. Du et al. [18] demonstrate that multi-agent debate improves factuality and reasoning through iterative multi-round convergence, where agents share responses and refine their positions over successive rounds. Liang et al. [19] extend this with explicit divergent thinking encouragement to counter “Degeneration-of-Thought”—the tendency of iterative debate to collapse into premature consensus. Chan et al. [20] propose ChatEval, a multi-agent debate framework for LLM evaluation with diverse role assignments.

IPCHA differs from these iterative convergence approaches architecturally: it implements a single-pass asymmetric falsification pipeline rather than multi-round symmetric debate. This design reflects a deliberate tradeoff. Iterative debate (Du et al., Liang et al.) can explore complex argumentative depth through successive refinement, but faces Degeneration-of-Thought risk [19]—agents converge on a shared (potentially wrong) position after several rounds, producing false confidence. IPCHA’s single-pass design with enforced model diversity and authority grounding avoids convergence pathology at the cost of argumentative depth. Whether single-pass falsification or iterative convergence produces better verification outcomes is an empirical question we have not benchmarked; a direct comparison against MAD or ChatEval on standardised evaluation sets is needed.

Multi-agent verification. LLM-as-a-Judge frameworks [21] use one model to evaluate another. IPCHA extends this with mandatory model diversity, authority grounding, and the

trialectic structure.

Hallucination detection. SelfCheckGPT [22] proposes zero-resource hallucination detection via self-consistency. Rawte et al. [23] and Ji et al. [24] survey hallucination broadly. IPCHA differs in using structured adversarial challenge rather than self-consistency checks.

NLI-based factual consistency. NLI has been applied to factual consistency evaluation in several prior systems. AlignScore [1] uses a unified alignment function trained across NLI and other NLP tasks to evaluate factual consistency of generated text. SummaC [2] operationalises consistency by checking whether each generated sentence is entailed by the source document and not contradicted. IPCHA differs from these systems in three respects: (1) it applies NLI within a *mandatory adversarial pipeline* rather than as a post-hoc scoring metric; (2) the NLI component scores structured claim–evidence pairs produced by the trialectic, not free-form generated text against a source; and (3) the scoring is embedded in a production multi-agent sycophancy defense system with authority grounding, model diversity enforcement, and fail-closed architecture. The NLI technique itself is not novel; the contribution is its integration into a structurally adversarial verification pipeline.

Natural Language Inference models. DeBERTa [12] provides the NLI backbone for our scoring upgrade.

11 Conclusion

The Ipcha Mistabra Protocol elevates sycophancy defense from a model-internal property to an architectural constraint. By institutionalising structured adversarial verification as a mandatory pipeline middleware, the protocol ensures that every claim faces adversarial challenge from an independent agent grounded in authoritative sources.

Our empirical evaluation on a 110-case synthetic corpus demonstrates that NLI-based scoring provides significant advantages over both TF-IDF and sentence-embedding baselines:

- **NLI classification adds value beyond better encoders:** NLI vs SBERT produces a large effect (Cohen’s $d = -0.80$, $p < 0.001$), confirming the advantage comes from entailment/contradiction classification, not merely from using transformer embeddings.
- **Score separation is the primary contribution:** NLI provides $4.1\times$ wider separation than TF-IDF and significantly different distributions from SBERT, giving higher confidence margins on borderline cases.
- **Classification accuracy is uniformly high** (98–100% for all methods) but not statistically differentiable at $n = 110$ (McNemar $p > 0.05$), and the perfect NLI accuracy reflects synthetic corpus simplicity rather than adversarial robustness—we do not claim accuracy superiority.
- **Weight calibration is robust** (point-biserial $r = 0.896$, stable across the full search range).

These results validate the NLI upgrade as a meaningful engineering improvement on controlled inputs. They do not establish production readiness: the same DeBERTa model achieves approximately 49.5% on ANLI Round 3 (an adversarially constructed benchmark), confirming that our synthetic corpus is substantially easier than real adversarial inputs. The weight calibration (Section 6.3) is exploratory due to circular data usage. Production deployment requires domain-specific calibration on substantially larger corpora ($n \geq 500$) with independent annotation and proper train/test separation.

Beyond the scoring improvement, the protocol’s architectural contributions—fail-closed coherence validation, sycophancy monitoring, denial-of-wallet defense, and authority grounding—provide defense-in-depth that no single-model alignment technique can offer. However, several load-bearing assumptions remain unverified: model-family independence is asserted rather than demonstrated, the audit infrastructure requires immutability guarantees not yet implemented,

and the authority document pipeline lacks integrity controls. The organisational dynamics around bypass mechanisms create predictable pressure toward the protocol becoming theatrical rather than functional.

We do not claim to solve the sycophancy problem. What the protocol provides is a *structural guarantee of adversarial challenge*—a necessary but not sufficient condition for epistemic integrity. The core architectural insight is sound; the implementation gap between current state and production readiness is substantial and honestly acknowledged.

Data Availability and Reproducibility

The IPCHA protocol is implemented and deployed as an internal research prototype within the nyxCore platform, a private multi-tenant dashboard system. The current deployment serves as an internal pilot for the author’s own workflows; it has not undergone independent security audit or compliance certification, and the production-readiness gaps identified in Section 8 (audit log immutability, authority document integrity, bypass dynamics) remain unresolved. We use the term “deployed” to indicate functional integration, not compliance-ready production status.

The IPCHA verification sidecar, NLI microservice, evaluation corpus (110 cases), and scoring scripts are available at <https://github.com/mrwind-up-bird/ipcha-mistabra> (release v1.0.0) under the Apache 2.0 license. The nyxCore platform, which provides the dashboard and workflow integration, is a proprietary system not required for reproducing the evaluation results. The ONNX-exported DeBERTa model is publicly available on HuggingFace (MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli).

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